What a reward model can show depends on who stopped the episode: failure judges and monitors on a low-cost arm

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Abstract

Reward models that score robot rollouts are also used as failure detectors, and the baseline they must beat depends on who stopped the episode. On 2,047 public SO-101 rollouts from 7 policies, where runs end at a time limit, plain episode length separates failure from success at AUROC 0.942 within policy — and still 0.851 when only half of the longest run has been observed: failed runs are slow, not timed out. Both zero-shot reward models we test score below that baseline (the best reaches 0.821 within policy). On our own operator-stopped rollouts the regime flips: length falls to 0.56, neither reward model beats the policy-identity null, and what works is a judge at rest — a robot-trained 27B judge passes its pre-registered bar, though its operating threshold is in-sample, and every mid-episode monitor we tested misses the bar. We recommend reporting a length baseline, within policy, and the dataset's stopping rule.

1 Introduction

A reward model that scores robot rollouts is also a failure detector, and papers evaluate it as one. Any such evaluation has a hidden variable: the dataset's *stopping rule* — who decides that an episode is over. When a benchmark stops failed runs at a time limit, how long an episode ran encodes its outcome, and a detector should be read against the baselines the stopping rule hands out for free: elapsed length, and the identity of the policy that ran. When an operator stops each run by judgment, those baselines collapse, and the task changes. None of the reward-model papers in our library reports how far plain episode length gets on the same data. We measure it on a public SO-101 benchmark [12] and on our own rollouts, with the length analyses frozen before the numbers, and set every learned judge and monitor we ran against it. Contributions: (i) a pre-registered length analysis of 2,047 public rollouts, pooled, within policy and truncated in time; (ii) the same analysis for two zero-shot reward models [5, 1], with the stopping-rule baselines beside them; (iii) judge and monitor gates on 88 operator-stopped rollouts of our own, with every in-sample threshold disclosed.

2 Data and labels

ArmnetBench v0.1 [12] (SO-101, 8 tasks, 7 policies): 2,047 policy rollouts with a success or failure class (515 successes, 1,532 failures); teleoperated episodes excluded. One task (pushing a Displayport cable, 270 rollouts) has rollouts of a single class and so no per-task AUROC. Our rollouts: 88 labelled bowls-task episodes on one SO-101 arm and a Jetson Orin NX, and the 50-episode subset (42 successes, 8 failures) every monitor gate used. Labels are the operator's, not blind to policy.

3 The length baseline

Table 1 carries the numbers. Pooled over policies, length gives 0.938 (CI 0.925–0.951); within each policy it ranges 0.91–0.96, pair-weighted 0.942 (CI 0.927–0.956): the baseline is not the policy identity (Fig. 1). Robometer-4B [5], run zero-shot as its authors ship it, gives 0.818 pooled (CI 0.790–0.843) and 0.821 within policy (pair-weighted over the per-policy scores, which span 0.79–0.87; its per-episode scores were not stored, so these carry no episode-bootstrap CI). The protocol fixed Robometer's primary score as its success probability; read descriptively from the same run, its last-progress head does better, 0.838 pooled, but stays below length. TOPReward [1] (Qwen3-VL-8B [11], no training; the primary score, fixed before any score, is the paper's success score) gives 0.653 pooled (CI 0.623–0.684) and 0.661 within policy (CI 0.631–0.692), below length by -0.31 to -0.25 (paired 95 % CI); its full-video Yes/No component alone reaches 0.678 within policy, descriptively.

*Code, result files and the checks that regenerate every number: github.com/ravediamond/paper-reward-models-stopping-rule. The frozen protocols are copied verbatim under repro/; the bar commits named in the paper belong to the authors' research repository ravediamond/road-to-robotics.

signal	pooled	within policy
episode length	0.938 (0.925–0.951)	0.942 (0.927–0.956)
length seen up to 50 % of the longest run	—	0.851 (0.829–0.873)
Robometer-4B, success probability	0.818 (0.790–0.843)	0.821 (per-policy 0.79–0.87)
Robometer-4B, last progress (descriptive)	0.838 (no CI)	—
TOPReward, success score	0.653 (0.623–0.684)	0.661 (0.631–0.692)
TOPReward, Yes/No component (descriptive)	—	0.678

Table 1: ArmnetBench, failure AUROC with 95 % episode-bootstrap CIs. The Robometer within-policy entry is pair-weighted over its per-policy block and carries the per-policy range instead of a CI (per-episode scores not stored); the last-progress entry is descriptive (the protocol's primary score was the success probability).

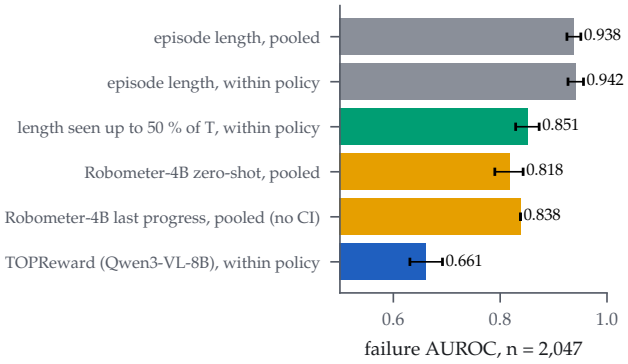


Figure 1: Failure AUROC on ArmnetBench (95 % CIs where available): episode length against two zero-shot reward models (Robometer-4B, TOPReward with Qwen3-VL-8B). The last-progress bar is descriptive and carries no CI. The Displayport-cable task contributes rollouts of one class only and has no per-task AUROC.

4 Is it the timeout?

Only 8.7 % of failures run to the task's longest episode. Observing length only up to half of that, when 88 % of failures but 19 % of successes are still running, gives 0.851 within policy (CI 0.829–0.873; Fig. 2; TOPReward on frames up to the same cut: 0.667): failed rollouts are slow, not timed out. Both pre-registered readings hold: length is a within-policy null, and it is more than the timeout. On our own operator-stopped rollouts the same baseline is at chance (0.56 on the 69; 0.52 on all 88). Neither reward model beats the policy-identity null there: TOPReward 0.82 (CI 0.70–0.93) against 0.87 for policy identity alone, and 0.78 on the 50 episodes without TurboVLA [16] (17 of the 25 failures in the 69 come from TurboVLA); Robometer 0.577 against 0.878 on 88. The baseline depends on how episodes end, which is why it must be reported per dataset; what works on operator-stopped runs is the judge at rest (§5).

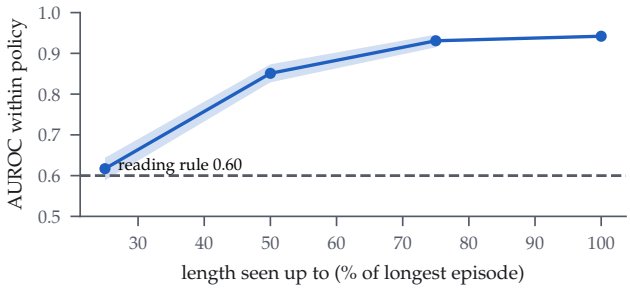


Figure 2: Length observed up to a fraction of the task's longest episode, AUROC within policy (band: 95 % CI).

5 Judges and monitors on our rollouts

Table 2 carries every judge and monitor with the same triple: threshold-free AUROC where defined, failures caught and successes flagged at the operating point. Every operating threshold of the judges at rest is chosen in-sample on the same 50 episodes; the AUROCs are not.

At rest. A deterministic top-camera check (colour masks and geometry, no learning) is right on 94.3 % of judged episodes with 0 false positives (Wilson 0–13 64-frame two-view clip and NVIDIA's prompt, Flora-Rewarder-27B [8] reaches a threshold-free AUROC of 1.00 (bootstrap CI 1.00–1.00); at its in-sample operating point it flags 8 of 8 failures at 2 of 42 successes (Wilson on the false alarms 1–16 bar. Its base model Qwen3.5-27B [10] reaches 0.91 (0.78–1.00; 7/8 at 2/42) on the same input at its own in-sample threshold; Qwen3.5-9B [10] reaches 0.75 (7/8 at 16/42) and Gemini 3.8 Flash [2] 0.96 (8/8 at 3/42, one false alarm over the bar): neither passes. The input format, not the robot fine-tune, decided the result.

Mid-episode. A Gemini sliding-window judge catches 0 of 8 failures at 1 of 42; the surprise of a separate world model, trained in the LeWorldModel recipe [7], catches 2 of 8 at 2 of 42 (AUROC 0.86, Wilson 7–59 (AUROC 0.30). An action-conditioned risk head on the world model's latent reached 0.93 in a first version that read elapsed time, and that result is withdrawn; the pre-registered second version, without time or length, catches 3 of 8 at 5 of 42 (AUROC 0.79 against 0.71 with shuffled actions; Wilson 14–69 bar of at least 5 of 8 failures at no more than 1 of 20 false interventions (Fig. 3).

6 Related work

Reward models are increasingly proposed as zero-shot failure detectors for manipulation: Robometer [5] scales a general-purpose reward model by trajectory comparisons, and TOPReward [1] reads token probabilities of a video-language model as hidden rewards; a recent survey of verifiers for robot policies [14] maps the wider family. Mid-episode failure detection is its own line:

method	kind	AUROC	caught / 8	flagged / 42	threshold
colour judge (no learning)	at rest	— (94.3 % right, 0 FP, 5 FN)	—	—	fixed rule
Flora-Rewarder-27B	at rest	1.00 (1.00–1.00)	8	2	in-sample
Qwen3.5-27B, same input	at rest	0.91 (0.78–1.00)	7	2	in-sample
Qwen3.5-9B, same input	at rest	0.75	7	16	in-sample
Gemini 3.8 Flash, same input	at rest	0.96	8	3	in-sample
Gemini sliding windows	monitor	—	0	1	fixed before scoring
world-model surprise	monitor	0.86	2	2	fixed before scoring
world-model latent drift	monitor	0.30	0	2	fixed before scoring
risk head, logistic (v2)	monitor	0.79	3	5	5-fold by episode
risk head, MLP (v2)	monitor	—	6	13	5-fold by episode

Table 2: Judges and monitors on 50 own operator-stopped rollouts (42 successes, 8 failures). Judges score the finished episode; monitors score mid-episode windows against the monitor bar ($\geq 5/8$ caught at $\leq 1/20$ false interventions). "In-sample" marks an operating threshold chosen on these same 50 episodes.

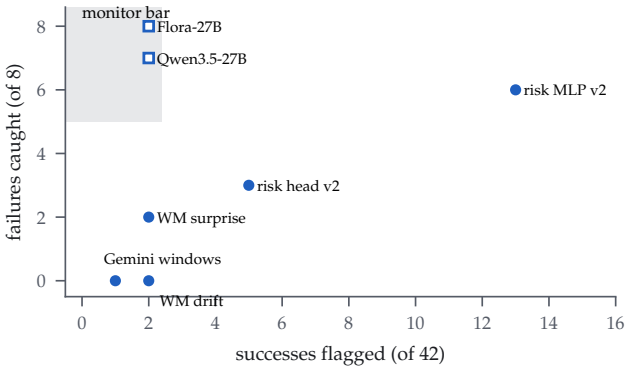


Figure 3: Fifty own rollouts: failures caught (of 8) against successes flagged (of 42). Circles: mid-episode monitors; shaded: their bar ($\geq 5/8$ at $\leq 1/20$). Squares: judges at rest (one score per finished episode) at their in-sample operating point; the monitor bar does not apply to them.

FAIL-Detect [17] and SAFE [3] learn uncertainty- and feature-based failure scores for imitation and VLA policies, FARM [9] reads failure signals from the internal states of a frozen world model, ReAP [6] predicts action-conditioned risk, VLA-Corrector [13] verifies stages to drive prompt-based recovery, Spotter [4] pairs the embodied model with a reflecting VLM, and REMORA [15] tracks task progress through errors and recovery with a recurrent distributional model. Benchmarks on the same low-cost arm — ArmnetBench [12] and a standardized SO-101 VLA benchmark [18] — evaluate policies; none of these works reports the elapsed-length baseline under the dataset's stopping rule, which is the comparison this paper adds.

7 Limitations

Two task families (ArmnetBench's eight tasks, our bowls task); our labels are the operator's and not blind; the risk head is trained and tested on the same 50 episodes (5-fold by episode: episodes do not leak, tasks and policies do); Robometer's within-policy and truncation analyses carry no episode-bootstrap CI and no truncation column because its per-episode scores were not stored; TOPReward's prefix count, frames per query and Yes/No wording are not given in its paper and are ours; the operating points of Flora-Rewarder-27B, Qwen3.5-27B and Gemini 3.8 Flash are chosen in-sample on the 50 episodes they are scored on (their AUROCs are not); 17 of the 25 failures in our 69 rollouts are TurboVLA's, so on our set any score that reads the policy gets the identity null for free; the timeout is approximated by each task's longest episode.

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analysis	repro folder	expected files	protocol
length baseline, truncation	repro/length-null	length_armnetbench.json, counts_ci.json	LENGTH_PROTOCOL.md, frozen 1a69f30
Robometer on ArmnetBench	repro/robometer-judge	g9a_ab_complete.json	PROTOCOL.md (primary score fixed there)
Robometer and length on own rollouts	repro/robometer-judge	g9a_own.json	PROTOCOL.md
colour judge	repro/robometer-judge	g9c3_test_results.json	PROTOCOL.md
TOPReward, both datasets	repro/topreward	topreward.json, own_rollout_labels.csv	TOPREWARD_PROTOCOL.md, 6241570
judges at rest, input-format gate	repro/flora-input-format	g_flora_summary.json, g_flora_abl_summary.json	PROTOCOL.md
window judge, world-model monitors	repro/six-monitors	judge_tau.json, w1a_summary.json	PROTOCOL.md
action-conditioned risk head	repro/reap	monitors.json, monitors_v2.json	PROTOCOL.md

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Appendix: from plain name to analysis

Every number is recomputable from the repository; this table maps each analysis to its files.